

Scalability & Future Plan

SKILL WALLET — A SmartBridge Product

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Local Memory Training Bound	Model cannot train on datasets larger than local RAM capacity, restricting retraining frequency as data grows.	Medium
2	Static Threshold Evaluation	Credit classification risk boundaries remain fixed, failing to adjust to fluctuating real-world macroeconomic shifts automatically.	High
3	Single Ingestion Stream	The system handles single-form input data, limiting bulk credit batch approval capabilities for banking partners.	Low

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local user session via lightweight web framework.	Migrate application endpoints to cloud architectures using container microservices (Docker/Kubernetes) to auto-scale on traffic load.

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2	Data Storage	Local system flat CSV file storage.	Integrate cloud-hosted relational DBs (PostgreSQL) paired with secure data warehouses for historical applicant audits.
3	Performance	Synchronous compute thread execution on local CPU bounds.	Utilize asynchronous task processing queues and GPU-accelerated environments for complex retraining model instances.
4	Security	Unencrypted local variable exchanges.	Enforce end-to-end data transit encryption via HTTPS, parameter masking for private financial data, and OAuth2 institutional authentication.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Automated MLOps Drift & Dynamic Threshold Tracking Pipeline Integration.	Q3 2026	Maintains inference consistency against evolving consumer spending changes.
Phase 3	Bulk Batch PDF/CSV Application Upload Processing Capability.	Q4 2026	Enables commercial banking institutions to evaluate entire loan portfolios concurrently.
Phase 4	Alternate Alternative Credit-Data Scoring Models (Utility/Rent profiles integration).	Q1 2027	Boosts prediction fidelity for demographics lacking established banking score histories.